Journal Pre-proof



Diversification in sight? A macroeconomic assessment of Saudi Arabia's vision 2030

Flavien Moreau, Zamid Aligishiev

PII: S2110-7017(24)00061-1
DOI: https://doi.org/10.1016/j.inteco.2024.100538
Reference: INTECO 100538

To appear in: *International Economics*

Please cite this article as: F. Moreau and Z. Aligishiev, Diversification in sight? A macroeconomic assessment of Saudi Arabia's vision 2030. *International Economics* (2024), doi: https://doi.org/10.1016/j.inteco.2024.100538.

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2024 Published by Elsevier B.V. on behalf of CEPII (Centre d'Etudes Prospectives et d'Informations Internationales), a center for research and expertise on the world economy.

Journal Pre-proof

Diversification in Sight? A Macroeconomic Assessment of Saudi Arabia's Vision 2030 *

Flavien Moreau, Zamid Alighishiev

August 17, 2024

Abstract

Saudi Arabia, currently the world's largest oil exporter, embarked since 2021 on a large-scale National Investment Strategy (NIS) designed to lift potential GDP growth by diversifying the economy. This article describes the strategy and is the first paper to quantifies its impact on growth using a dynamic general equilibrium model. It also decomposes this impact into two main channels: the direct impact of the investment push and the impact of complementary reforms. We find that the overall strategy—when supported by appropriate fiscal measures, labor supply reform, and higher public sector efficiency— could boost potential non-oil growth by 4.8 percentage points to about 8.8 percent in the medium term. We also assess the growth dividends of a wider range of alternative scenarios around the NIS baseline.

KEYWORDS: Economic diversification, Public Investment, Potential GDP growth, Sovereign wealth funds, DSGE

JEL CODES: E1, E2, F4, Q3, H5

*Flavien Moreau: International Monetary Fund. E-mail: fmoreau@imf.org; Zamid Aligishiev: International Monetary Fund. E-mail: zaligishiev@imf.org. The views expressed in this article are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management. A summary of this article's findings was presented in Saudi Arabia's 2022 Article IV report. Authors would like to thank Cian Ruane for inputs, suggestions, and comments; Giovanni Melina for useful explanations about the DIGNAR model; and Amine Mati, Fozan Fareed, Jerome Vacher, Sidra Rehman for helpful discussions. We would also like to thank Mohamed Alrashed and Saudi authorities for informative discussions. We thank Tewodaj Mogues for sharing data on public investment efficiency.

Journal Pre-proof

1 Introduction

Natural endowments shape the structure of comparative advantages of resource-rich economies. Few resource-rich countries have managed to diversify successfully, that is, before they were forced to, either because the resource ran out or became unprofitable to extract. Successful cases – usually small open economies - have often proceeded by saving a large fraction of the proceeds into sovereign wealth funds and developing their financial and real estate sectors. In contrast, cases of mismanaged resource proceed abound.

Against this background, Saudi Arabia paints a singular picture. An oil-based economy, with oil exports equivalent to about 40 percent of GDP and more than two thirds of fiscal revenues (Vacher, 2022), the Kingdom is also a G20 member and a large economy – with a population of about 37 million in 2022. It possesses a well-endowed sovereign wealth fund, the Public Investment Fund (PIF), whose balance sheet has considerably expanded over the past few years. Since 2021, the Kingdom has embarked on an ambitious plan of structural economic transformation called "Vision 2030", in which PIF is given a central role: to catalyze private funding and to spearhead new sectors such as, among others, hospitality, digital services, and sports – well beyond the Kingdom's traditional commodity exports (Ministry of Investment, 2022).

The scale and pace of "Vision 2030" are unparalleled, leaving very few natural yardsticks. In addition, macroeconomic studies of Gulf countries' growth strategies have so far been limited and did not produce internally consistent assessments such as those based on general equilibrium models.[1] This article fills this gap by proposing a macroeconomic analysis of Saudi Arabia's structural transformation strategy. To our knowledge, it is the first general equilibrium analysis of Vision 2030.

We quantify the growth dividends from Vision 2030 through the lens of a dy-

[1] One important and notable exception is Cherif and Hasanov (2016)

Journal Pre-proof

namic general equilibrium model tailored to analyze the implications on the real, fiscal, external sectors of large investment scale-ups in commodity-rich countries. In particular, the model possesses three sectors, a resource-based sector, a tradeable sector, and a non-tradeable sector. As a result, it allows to account for the impact on trade as the investment plan is implemented. We also decompose these dividends into two main channels: the direct impact of the investment push and the impact of complementary reforms.

We model several alternative implementation scenarios, reflecting possible paths along which the investment strategy could unfold and be combined with packages of reforms. This approach allows to account for possible implementation risks which partly explain why, despite previous attempts at diversification, the share of oil activities to GDP has remained around forty percent.[2] Under the full implementation scenario, we find that the initiatives in the NIS can substantially raise non-oil potential growth, by up to 4.8 percentage points, conditional, in particular, to the ability to maintain the ambitious investment path prescribed by the NIS. We then complement that analysis by examining alternative scenarios with different investment paths and fiscal multipliers. It is important to note that several important sources of uncertainty remain, including the trajectory of oil prices as well as geopolitical factors. As this study is mainly focused on modelling internal factors, we leave the analysis and integration in the model of these sources of aggregate uncertainty for further research.

The structure of the paper is the following. We first briefly review the related literature, then provide a broad description of Saudi's National Investment Strategy. The modeling approach is described in the fourth section. Section 5 presents the main findings and Section 6 provides additional robustness checks, in particular concerning the implied size of the multipliers. Section 7 concludes.

[2]See for instance Albassam (2015), Al Naimi (2022), Hvdit (2013), and Sweidan (2021).

Journal Pre-proof

2 Related Literature

The paper speaks to a large literature on export-oriented growth accelerations, which seeks to identify the factors driving spells of elevated and sustained growth rates, as well as a nascent literature on economic diversification in the Middle East. Most of the literature has focused on the growth episodes in East Asian countries, where several countries like Korea or Singapore have managed to escape the "middle income trap" (Cherif and Hasanov, 2019). Examples of growth accelerations in the Middle East have been limited to a handful of episodes – in Jordan, Syria, or Tunisia –. While growth accelerations episodes have been associated with increases in investment and trade in particular, many instances of reform failed to produce growth accelerations, which are also influenced by a host of other idiosyncratic factors (Hausmann et al., 2005). However, leadership changes and large-scale economic reforms are a predictor of growth accelerations (Hausmann et al., 2005), therefore making Saudi Arabia a natural candidate for the study of growth accelerations.

Economic growth among the Gulf Cooperation Council (GCC) countries – of which Saudi Arabia is a member – has been historically tied to their exploitation of large oil and gas reserves. However, natural resource abundance has so far failed to translate into sustainable economic growth (Sachs and Warner (1995), van der Ploeg (2011)). This natural resource "curse" has been attributed to several factors, including the volatility of commodity prices, with the upswing and downswings of commodity prices generating large fluctuations in the balance of payments and fiscal accounts. Overoptimistic forecasts of the path of oil prices have led to excessive borrowing followed by growth stagnation as prices collapsed (Manzano and Rigobon, 2007). Declining public investment efficiency (van der Ploeg, 2012) and the presence of bottleneck have also led to under-whelming growth spurts, especially in resource-rich countries (Berg et al., 2013). We stress that point in our analysis by examining the impact of different levels of public investment efficiency.

Our article also contributes to a nascent macroeconomic literature on Saudi

Arabia. The Kingdom has been marked by more than four decades of efforts to diversify the economy away from oil dependency (Al Rajhi et al., 2012). Time series analyses have studied the differentiated impacts of financial development on the growth of the oil and non-oil sectors (Samargandi et al., 2014) or analyzed whether oil price collapses have been associated with structural breaks in growth rates (Jawadi and Ftiti, 2019). Sweidan and Elbargathi (2023) and Pierru and Matar (2014) also emphasize the role of oil price uncertainty on Saudi's economy, and the propensity to boom bust cycles. Jawadi and Ftiti (2019) suggests that diversification in the form of equity investments in non-oil sectors, consistent with Vision 2030, could help sustain growth in the medium run. While these time series studies shed light on important linkages between Saudi's economy and oil prices, they are not suitable to study the channels through which the NIS is to affect the various non-oil sectors.

Our article is the first general equilibrium assessment of the investment strategy at the heart Vision 2030, to the best of our knowledge. Assessments of KSA's diversification plans have been mostly qualitative. For instance, Albassam (2015) provides a systematic, qualitative assessment of ten development plans undertaken by Saudi Arabia's Ministry of Economy and Planning, showing that oil has remained the dominant engine of the economy, despite diversification being an important goal of most these plans. Guendouz and Ouassaf (2020) use a regression analysis to identify correlates of economic diversification, as measured by a Normalized Herfindahl Hirschman Index. Closest to our paper, Blazquez et al. (2021) use a DSGE model to examine the impact of some policy changes – a VAT introduction and energy reforms – that were introduced before the NIS was unveiled. Therefore they do not model the multi-billion public investment ramp-up on which this paper focuses. In spite of this, Blazquez et al. (2021) find large welfare gains from these policies and in an increase of GDP of up to 5 percent.

To quantitatively assess the macroeconomic implications of NIS for Saudi Arabia's economy, we extend the small open economy model in Melina et al. (2016).

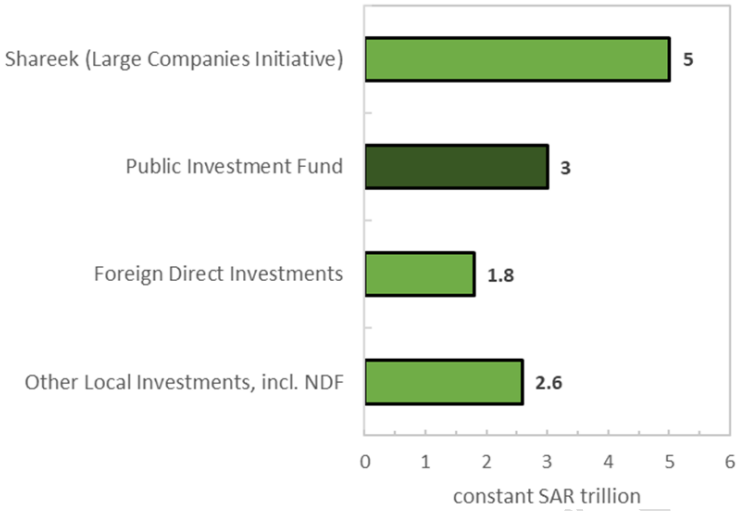
This variant, denoted the DIGNAR-19 model (Aligishiev et al., 2021), accommodates exogenous shifts in the disutility of labor and a richer external sector. We further modify this model to account for the fact that only a fraction of the sovereign wealth fund is invested abroad to reflect the lack of exchange-rate-driven valuation changes in its domestic component. This paper fits into a broader literature on applying Dynamic General Equilibrium models to evaluate the macroeconomic impacts of and identify policy lessons from large investment scale ups in resource-rich economies. Some notable examples include Berg et al. (2013), Melina et al. (2016), and Delechat et al. (2015).

Finally, our work has important policy implications, and touches upon a rapidly expanding literature examining diversification strategies and industrial policies. While most of the attention has been directed to the East Asian experience, arguably successful in several respects, there has been more limited studies of diversification strategies in Middle Eastern countries. One notable exception is Cherif and Hasanov (2016) who argue that to achieve sustainable growth, GCC countries need to develop dynamic non-oil tradable sectors and propose several options to alleviate the market failures. More recently, Cherif et al. (2024) emphasize that even a low probability of an accelerated energy transition would entail enormous costs for oil-rich countries and suggest that ambitious industrial policies can be transformational as long as they adopt a risk-based approach.

3 Saudi Arabia's National Investment Strategy

Under "Vision 2030", the authorities have set out to ramp up both public and private investment, attract FDI and diversify the economy. The NIS is under the supervision of the Council of Economic and Development Affairs (CEDA) and is supported by four pillars, namely, 1) developing new investment opportunities, in strategic sectors 2) broadening the investment base, 3) removing financial barriers to private sector investment, and 4) improve the competitiveness of the business

Figure 1: Cumulative Investment Targets (2021-2030)

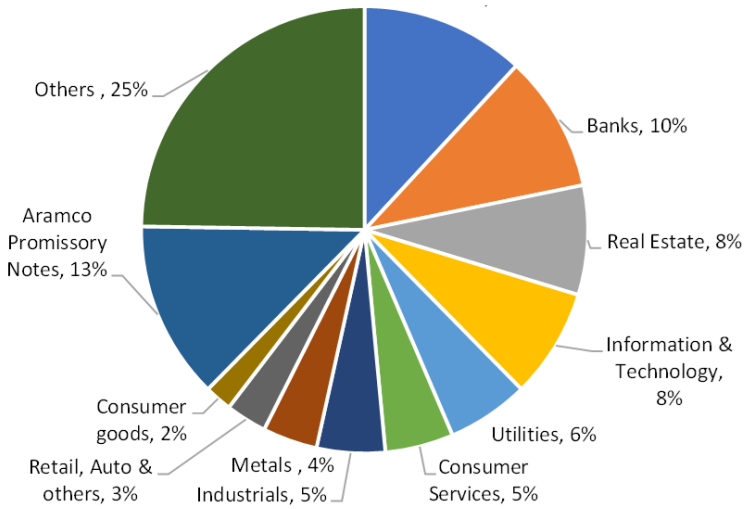


Source: Saudi Arabia Authorities

environment. These objectives are backed by more than 40 initiatives. After a first phase focused on overhauling the regulatory framework, efforts are now turning to tackling structural challenges and attracting investments. While diversification has been a recurrent focus of Saudi Arabia's 10 development plans since 1970 (Albassam (2015)), the NIS stands out by its ambition and the degree of precision in which the goals are formulated.

Under the NIS, Gross Fixed Capital Formation (GFCF) is expected to grow by 13 percent per year until 2030. In nominal terms, this corresponds to a total of SAR 12.4 Tr cumulative new investments in the Saudi Economy by 2030, or about twice what would be invested if investments were growing at the historical average. In support of this strategy, the Public Investment Fund (PIF), Saudi Arabia's Sovereign Wealth Fund, is expected to invest at least SAR 150 bn per year in the Saudi economy (Figure 8). Important contributions are also expected from the National Development Fund (NDF), whose consolidated assets reached about 17 percent of GDP in 2019. In addition, the NIS seeks to unlock private investment through the Shareek program, or Large Company Initiative, expected to generate SAR 5 Tr in investments. At the current juncture, diversifying the Saudi

Figure 2: Figure 2. NIS and Strategic Sectors



Source: Moody's

economy remains a challenge, as around 60 percent of current annual investments are concentrated in the hydrocarbon sector, as well as more than half of the FDI.[3]

The NIS seeks to stimulate private investments, including in key strategic sectors, which the PIF has been tasked to help develop. These 13 strategic sectors include: Aerospace and Defense; Automotive; Transport and Logistics; Food and Agriculture; Construction; Entertainment, Leisure, and sports; Financial services; Real Estate, Utilities and Renewables; Metals and Mining; Health care; Consumer Goods and Retail; and Telecom, Media, and Technology. As of December 2020, some of these sectors were already well represented among PIF total assets holdings. While the catalytical effect on private sector is expected to vary, evidence suggests potential strong response from private companies in sectors such as Entertainment, Health, and Renewables (IMF, 2020).

The overall strategy shares some parallels with successful strategies implemented in fast-growing economies that have escaped the middle-income trap, such as Korea or Singapore. Both countries relied on policy mixes that favored large investments,

[3]Inward FDI jumped from SAR 20.3 bn in 2020 to SAR 72.4 bn in 2021, of which SAR 46.5 bn correspond to the sale of a 49 percent equity stake in Aramco Oil Pipelines to a consortium led by EIG.

Journal Pre-proof

in particular in R&D, in order to promote technology upgrading and diffusion. In that respect, sectors where the country already has a strong comparative advantage, such as oil and petrochemicals for Saudi Arabia, can also help support vertical diversification and technological diffusion through other sectors. Other policies that shape incentives in favor of technical education and export sophistication play an important role (Cherif and Hasanov, 2019). While it is too early to take stock of the NIS's impact on innovation, such as the number of patents, evidence suggest that export sophistication has been increasing in the past years (Prasad et al., 2024).

Using the DIGNAR-19 model calibrated to the Saudi economy, we find that simulations of the initiatives undertaken under the NIS can substantially raise non-oil potential growth. However, the magnitude of the increase and its sustainability are subject to several uncertainties, in particular concerning the returns on capital, efficiency of investment expenditure, and the degree to which reforms will be successfully implemented and deliver the expected results. To quantify these uncertainties and highlight the role of the key assumptions, we formulate several scenarios and examine different calibrations of the model.

4 Model

We quantify growth dividends from reforms and public investment scale-up as well as the impact on private investment using simulations from an extension of the DIGNAR-19 model. DIGNAR-19 is a dynamic general equilibrium (DGE) model (see Aligishiev et al. (2021)) designed to deliver quantitative macroeconomic assessments and policy scenario analyses in open developing economies.[4] The framework abstracts from the full input-output structure of the economy and incorporates three sectors of production (nontraded goods, non-resource traded goods and traded nat-

[4]We build on the DIGNAR model by Melina et al. (2016) that accommodates the effects of the COVID-19 pandemic. Both DIGNAR and DIG models have gained a wide measure of acceptance for policy work. Over the past ten years, they have complemented the IMF and World Bank Debt Sustainability Framework (DSF) analysis, with over 65 country applications, in the context of Fund programs and surveillance work, and are now part of the IMF-WB Multipronged Approach for addressing debt vulnerabilities.

Journal Pre-proof

ural resources) as well as financially and non-financially constrained households.[5] The government has access to several fiscal instruments (productive and unproductive expenditures and various taxes) and both concessional and non-concessional debt, both domestic and external. This section highlights relevant features of the DGE model as well as the extensions made for the application to Saudi Arabia. For the full set of equations of the baseline model, the reader is referred to Melina et al. (2016).

It is important to note that the productivity gains from the NIS have an asymmetric effect on the non-resource tradable and non-tradable sectors in the model. This captures the varying dynamics between sectors such as tourism, transport and logistics, and industry (considered part of the tradable sector) and digital infrastructure, renewable energy, and healthcare (part of the non-tradable sector). The relative price of tradable goods following a productivity gain is more rigid—as a result of bias towards tradable goods in government purchases—therefore, a symmetric productivity improvement will increase the relative price of these goods and result in temporarily higher marginal products in this sector. This incentivizes resource reallocation, leading to higher investment expenditure and employment in the tradable goods sector. Over the long term, the two sectors are expected to converge.

The multiplier associated with the public investment program is determined endogenously by the magnitude of the investment surge, the source of financing, and the parameterization of the model. The model incorporates three main economic mechanisms for the analysis of public investment scale-ups: (i) a growth-investment nexus, (ii) a fiscal response required to maintain debt sustainability, (iii) and potential for both crowding-in and crowding-out of private investment. Public investment is assumed to be productive but facing diminishing returns. Higher public capital raises productivity of private factors and increases real output through:

[5]See Baqaee and Farhi (2019), Carvalho and Tahbaz-Salehi (2019) and Liu (2019) for recent advances in input-output macroeconomic modeling

$$\underbrace{y_{i,t}}_{\text{Output}} = \underbrace{A_{i,t}}_{TFP} \underbrace{\left(K_{t-1}^{G}\right)^{\alpha_g}}_{\text{Public capital}} \underbrace{\left(L_{i,t}\right)^{\alpha_N}\left(K_{i,t-1}\right)^{1-\alpha_N}}_{\text{Private factors of production}}, \text{ for } i \in \{T, NT\} \tag{1}$$

where $y_{i,t}$ is the real sectoral output; $A_{i,t}$ is the total factor productivity; $L_{i,t}$ is labor input; $K_{i,t}$ and K_t^G are private and public capital stocks respectively. $\alpha_g \in (0,1)$ and $\alpha_N \in (0,1)$ govern the return on public capital and the labor share in production, respectively. The level of public capital at any given year is a sum of the stock of capital in the previous year, net of depreciation, and the new effective public investment expenditure. Formally:

$$\underbrace{K_t^G}_{\text{New gov.capital stock}} = \underbrace{\left(1-\delta\right)K_{t-1}^G}_{\text{Old capital after depreciation}} + \underbrace{\epsilon I_t^G}_{\text{New investment after waste}} \tag{2}$$

where I_t^G is the total public investment expenditure; $\delta \in (0,1)$ is the depreciation rate; and $\epsilon \in (0,1]$ governs the efficiency of public investment. Higher investment expenditure translates into higher capital stock, which in turn increases the marginal product of private capital. Higher marginal product of capital incentivizes the private sector to match higher public expenditure with more private investments. The strength of this crowding in effect depends on the assumed size of the private investment adjustment costs:

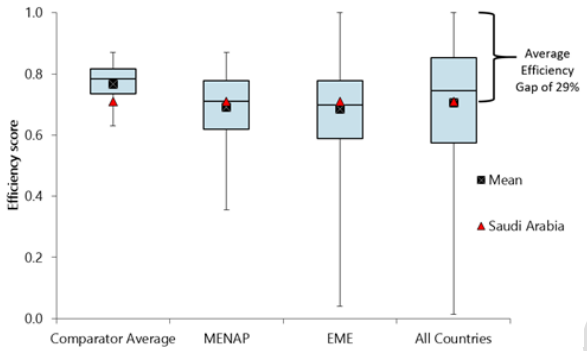
$$\underbrace{K_{i,t}}_{\text{New capital stock}} = \underbrace{1-\delta K_{i,t-1}}_{\text{Old capital after depreciation}} + \underbrace{I_{i,t}}_{\text{New investment}} \tag{3}$$

$$\underbrace{-\frac{\kappa}{2}\left(\frac{I_{i,t}}{I_{i,t-1}}-1\right)^2 I_{i,t}}_{\text{Adjustment cost}}, \text{ for i} \in \{T, NT\} \tag{4}$$

where I_t is private investment expenditure and $\kappa \in [0,1)$ governs the size of the adjustment cost. Higher values of α_g and ϵ increase the size of the investment multiplier, while higher values of κ decrease it.

Gains in public investments efficiency play an important role in the model. Pub-

Figure 3: Gains in Public Investment Efficiency



Source: IMF, Fiscal Affairs Department and Baum et al. (2020)

lic investment efficiency is the ability to translate a given level of spending into public goods, in particular economic and social infrastructure. On average, countries lose about one-third of their resources in the public investment process. As sources of inefficiencies are numerous and multidimensional, Baum et al. (2020) builds a hybrid indicator accounting for both volume and quality of infrastructure and compute an efficiency frontier. With an efficiency of 0.71 measured over the past decade, Saudi Arabia faces significant gains from closing its efficiency gap.[6] Higher efficiency of public investment, ϵ, translates a given amount of public investment expenditure into a higher public capital stock, which in turn delivers a larger increase in private investment.

Labor market reforms trigger a persistent shift in the labor supply curve. To model the anticipated increase in the labor supply stemming from the implementation of labor market reforms, we assume a fall in households' disutility of labor. Formally, labor supply of the two types of households (optimizers and rule-of-thumb) is given by equating the marginal disutility of supplying an extra unit of labor to the compensation received for it via wage income, subject to distortions arising from taxation:

[6]While there is no direct mapping between PIMA scores and the concept of public investment efficiency in the model (Gurara et al. (2019), they can provide a good proxy for understanding the level of efficiency relative to peers and feasible room for efficiency gains.

$$\underbrace{exp\left(\varepsilon_t^{LS}\right)\kappa_t^i\left(L_t^i\right)^{\psi^i}}_{\text{Disutility of labor}} = \underbrace{\frac{\left(1-\ \tau_t^L\right)}{\left(1+\tau_t^c\right)}}_{\text{Tax distortions}}\underbrace{W_t}_{\text{Wage}}\ \text{, for}\ \ i \in \{OPT, ROT\} \tag{5}$$

where L_t^i is the labor supply; W_t is the real wage; τ_t^c and τ_t^L are consumption and labor income tax rates, respectively; ψ^i is the inverse of the inter-temporal elasticity of substitution of the labor supply (Frisch elasticity); κ^i is the disutility weight of labor; and ε_t^{LS} is the labour supply shock. A permanent reduction in ε_t^{LS} implies an increase in the labor supply for both household types for any given wage rate.

To better fit the case of Saudi Arabia, we also update the external sector of the standard DIGNAR-19 model. The model considers a case of a small open economy with a flexible exchange rate regime. The Balance of Payments' dynamics is governed by the following two equations:

$$\underbrace{CAD_t}_{\text{Current account}} = \underbrace{\Delta D_t^C + \Delta B_t^* - \Delta f_t^F}_{\text{Financial account}} \tag{6}$$

$$\begin{aligned} CAD_t = & \underbrace{M_t\ - Y_t^O - P_t^{TH}X_t}_{Trade\ balance} \\ & + \underbrace{(1-div)*\Omega_t^O\ - \widetilde{r^*}f_{t-1}^F + \ r_{t-1}^{dc}D_{t-1}^C + r_{t-1}^*B_{t-1}^*}_{Primaryincome} \\ & + \quad \underbrace{REM_t}_{\text{Secondary income}} \end{aligned}$$

where CAD_t is the current account deficit; Δf_t^F is the change in the foreign component of the SWF; D_t^C and B_t^* are external commercial public debt and private foreign debt, respectively; M_t is imports; Y_t^O is exports of natural resources; $P_t^{TH}X_t$ is exports of non-resource tradable goods; $\widetilde{r^*}$, r_t^{dc}, and r_t^* are the risk free global rate, interest rate on external commercial debt and interest rate on private foreign debt, respectively; REM_t is net remittances. For simplicity, the Balance of Payments and fiscal variables are expressed in nominal foreign currency terms in the above.

Journal Pre-proof

Model simulations assume that the fiscal gap is closed through withdrawals from the sovereign wealth fund, with possibly different rates of withdrawal between the domestic and foreign components. The Saudi Arabia's SWF has two components: invested in domestic assets and invested in foreign ones. Formally:

$$\underbrace{f_t}_{SWF} = \underbrace{f_t^D}_{Domestic} + \underbrace{f_t^F}_{Foreign} \tag{7}$$

where f_t is the total value of the SWF and f_t^D is the value of the domestically invested component. Model simulations assume that the fiscal gap is closed through withdrawals from the sovereign wealth fund:

$$\underbrace{f_t - f_{t-1}}_{Change\ in\ the\ SWF} = \underbrace{OFB_t}_{Fiscal\ balance} \tag{8}$$

where OFB_t is the overall fiscal balance. The government may decide to leverage the domestic component or the foreign one, or both. This is governed by parameter κ^f:

$$\kappa^F f_t^D = \left(1 - \kappa^f\right) f_t^F \tag{9}$$

When $\kappa^f = 0$, only the foreign component of the SWF is used to finance the fiscal gap. The opposite holds for when $\kappa^f = 1$. It is worth noting that the impact of additional public investment on the overall balance is three-fold. Additional investments are above-the-line transactions; hence they reduce the overall balance as expenditures rise. Moreover, higher public investment expenditure tends to increase domestic interest rates as the return on private investments rise. Resulting higher interest payments on the stock of domestic debt reduce the fiscal balance, However, a wider tax base produced by the higher stock of public capital (through higher demand for private capital and employment, as well as an eventual increase in private consumption) supports the overall fiscal balance through higher tax receipts.

The DIGNAR-19 model has several important limitations. First, the model

does not endogenize foreign direct investment and hence mainly focuses on the impact NIS has on mobilizing domestic investment. It therefore does attempt to quantify important channels through which the NIS affects the financial account of the Balance of Payments. Second, the model does not have an explicit financial sector, nor does it include financial constraints. This modeling assumption means that private investment choices are mainly driven by the Tobin's Q and only face a single distortion in the form of investment adjustment costs.

5 Parametrization

The DIGNAR-19 model is calibrated for Saudi Arabia at an annual frequency using recent data capturing salient features of Saudi Arabia's economy. Table 1 reports the key parameter values necessary to pin down the initial steady state. Where possible, parameters are calibrated in line with data provided by the IMF country team. The initial rate of return on standard infrastructure investment, α_g,is set at 25 percent, approximately in the middle of the range of estimates identified by the World Bank (2010).[7] The initial efficiency of public investment, ϵ, is set at 70 percent in line with most recent country assessment. Table 1 highlights main parameters and initial values. Additional parameters are as in Aligishiev et al. (2021). Note that most parameters are close to ones used in Blazquez et al. (2021).

6 Results

We consider the following scenarios:

• *Baseline*. This scenario is the one aligned with the macroframework underpinning the June 2022 Article IV Consultations, which assumes that GFCF would keep growing at a rate of about 6 percent in nominal terms, with some partial implementation of the structural reforms under Vision 2030, delivering a medium-term

[7] In "Cost-Benefit Analysis in World Bank Projects" (2010), the Bank's Independent Evaluation Group identifies a 17 to 31 percent range.

Journal Pre-proof

Table 1: Calibrated Parameters and Initial Values for the Steady State

Definition	**Value**
Potential real GDP growth rate	3.2
Exports to GDP ratio	33.0
Imports to GDP ratio	16.9
Private investment to GDP ratio	13.3
Public investment to GDP ratio	12.1
Public consumption to GDP	20.0
Total public revenues to GDP ratio	40.0
Share of labor taxes in total public revenue	10.0
Share of consumption taxes in total public revenue	20.0
Share of natural resource revenues in total government revenues	57.3
Public debt to GDP ratio	25.2
Private foreign debt to GDP ratio	53.8
Value added of the natural resource sector	40.0
Real interest rate on domestic debt	1.0
Initial return on public capital	25.0
Public Investment efficiency	70.0
Depreciation rate of public capital	7.0
Investment adjustment costs	5.0

Note: Authors' calculations based on official statistics.

non-oil GDP growth of 4 percent.

• *Moderate NIS investment scale up with stronger reform implementation.* This scenario is considering a richer set of policy changes, with a larger number of reforms implemented, but with a more measured pace of investment increases. compared to the NIS. To be more precise, in this scenario, GFCF would grow at 9 percent on average on an annual basis after 2022. If sustained until 2030, this would be sufficient to cover about 90 percent of the investments initially planned under the NIS, given that the first few realized years have exceed initial projections. The reform scenario includes fiscal, labor, public investment, and regulatory reforms. Measures are assumed to be adopted in four areas: (1) a larger scale-up in combined public and PIF investment of 2.4 additional percent of GDP in 2022 and 5 percent of GDP in 2023-2026 ; (2) improvements in public investment efficiency is set to increase by 20 percentage points, following rationalization of procurements and other measure following PIMA guidelines (see Box 1); (3) fiscal reforms, including closing tax efficiency gap and rationalization of the high wage bill, including through strategic

workforce planning and review, yield a gradual improvement of the fiscal balance of 0.5 percent of GDP by 2026; (4) labor market policies, boosting labor force participation, including of Saudi women, increases labor supply by 2 percentage points by 2026 relative to the baseline scenario.

• *NIS pure investment scale up.* This scenario assumes that the NIS investment targets are fully met and follow the 13 percent annual growth rate specified in the NIS. Implementation of structural reforms is assumed to be the same as in the baseline. Therefore, this scenario isolates the differential impact on non-oil potential output of the large NIS scale-up relative to the baseline.

• *NIS investment scale up combined with reforms.* This scenario also assumes that NIS investment targets are fully met and that, additionally, all the policies envisaged in the reform scenario are implemented.

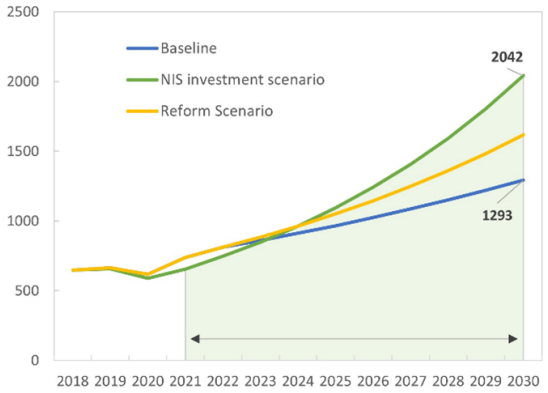
The scenario with full NIS investment scale-up combined with reforms could raise non-oil potential output growth to 8.8 percent in 2026, compared to 4 percent in the baseline scenario used in the current macro framework. Importantly, the impact from the pure investment scale-up itself would be hump-shaped, peaking in 2023 and slowly declining after that. In contrast, the reform package would build over time and help achieve long-lasting growth. Even if the full investment targets were not fully met, simulations show that a package of reforms coupled with a moderate investment scale-up would still boost non-oil potential growth at a gradual and sustained pace, reaching 7.7 percent in 2026 (Figure 4).

Our results indicate that the combined effect of the reform package could deliver a medium-term increase in non-oil real GDP growth of 4.8 percentage points relative to baseline. The growth dividends build over time as reform indicators improve (Figure 4). While the steep ramp up in public spending, account for most of the potential output gain in the first years of the simulation horizon, backloaded dividends from labor market reforms and from gains in public investment efficiency boosts non-oil growth in the medium run, by about 0.5 and 0.8 percentage points respectively compared to the baseline. In 2026, dividends from the investment ramp

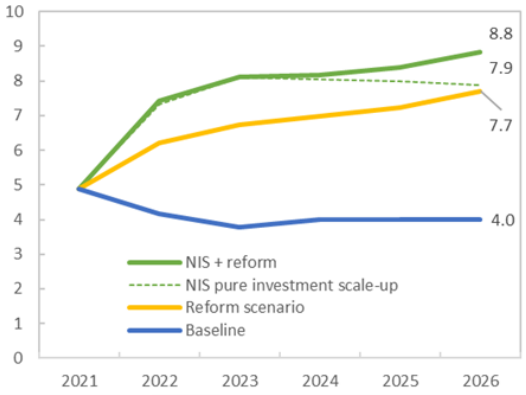
Journal Pre-proof

Figure 4: Raising Non-oil GDP

Investment Paths (SAR bn)



Non-oil GDP Growth Paths(Percent, y-o-y change)

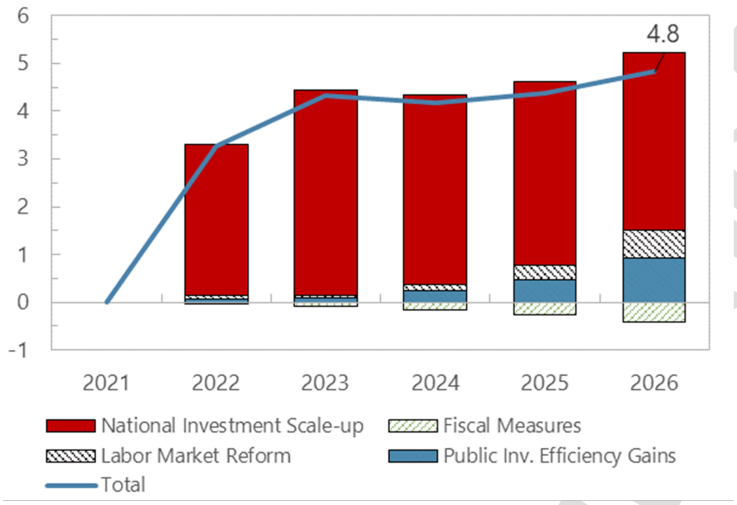


up amount to 2.7 points, with a modest drag of 0.4 percentage point from fiscal measures.

Our results indicate that the combined effect of the reform package could deliver a medium-term increase in non-oil real GDP growth of 4.8 percentage points relative to baseline. The growth dividends build over time as reform indicators improve (Figure 5). While the steep ramp up in public spending, account for most of the potential output gain in the first years of the simulation horizon, backloaded dividends from labor market reforms and from gains in public investment efficiency boosts non-oil growth in the medium run, by about 0.5 and 0.8 percentage points respectively compared to the baseline. In 2026, dividends from the investment ramp up amount to 2.7 points, with a modest drag of 0.4 percentage point from fiscal

Figure 5: Non-oil GDP Growth Dividends from Reforms

(Percentage points, relative to no-reform scenario)



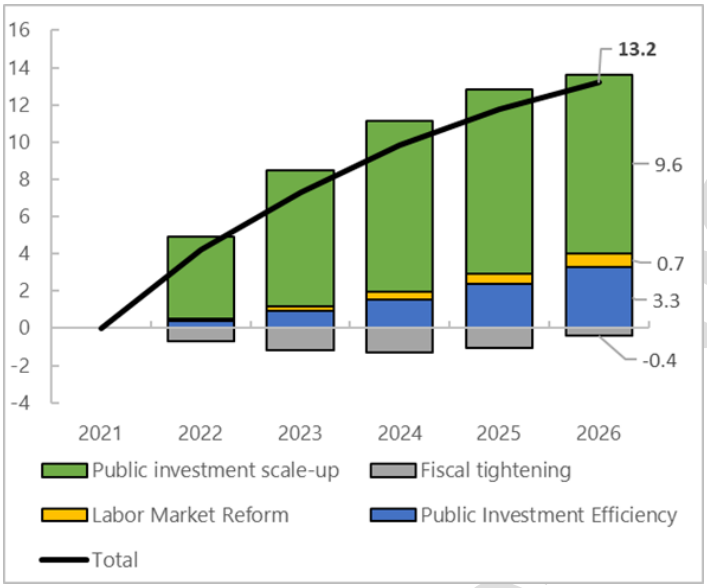
measures.

The growth gains implied by the reform scenario would help mitigate the risks related to an overreliance on oil revenues while catalyzing private investments. Private investments are expected to increase by up to 13.2 percentage point relative to the baseline (Figure 6). Simulations confirms the key role of the PIF's investments in driving additional private investments, while benefitting from positive spillovers from improvements on the labor market and in public investment efficiency. Under this scenario, and thanks to the modest fiscal consolidation, debt would stay at its current level and inflationary pressures on the real exchange rate would be contained, allowing the Saudi economy to rebalance towards a less oil-reliant economic model and generate more opportunities for the private sector.

The Vision 2030 target of reaching an investment share of GDP of 30 percent would be reached earlier than 2030 in the reform scenario. While our main simulations anticipate a more modest annual increase in gross fixed capital formation (GFCF) of about 9 percent per year in nominal terms than expected under the NIS, GFCF as a share of GDP would rise faster than in the NIS forecasts, reaching 29 percent of GDP in 2026 and coming close to the range maintained by high-growth

Figure 6: Impact on Private Investment

(Percentage points, relative to no-reform scenario)

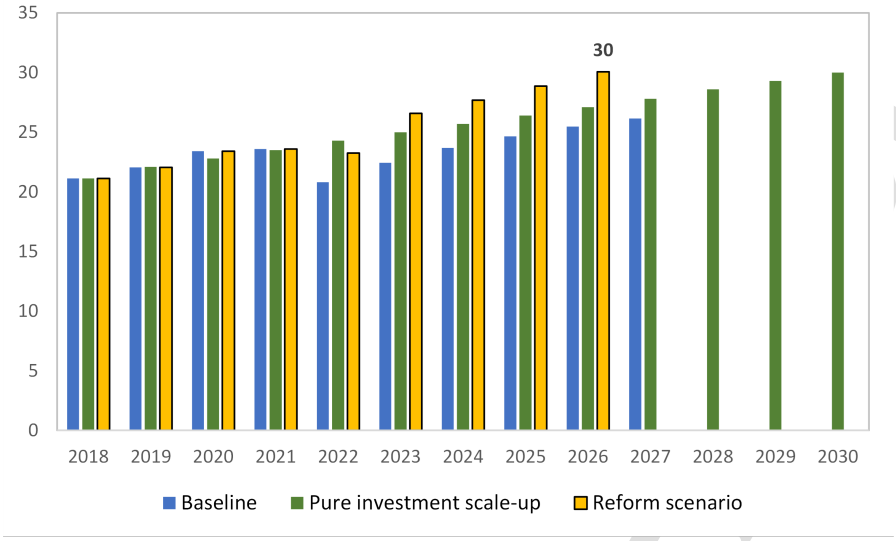


countries such as Korea or Singapore during the East Asian miracle. This faster pace is the result of discrepancies in investment multiplier assumptions, as, mechanically, lower multipliers result in higher GFCF shares for the same investment path.

From a quantitative standpoint, the size of the investment multipliers therefore play an important role in driving these results. A wide range of multipliers can be found in the literature, reflecting a large heterogeneity in both countries' characteristics, the types of investment plans, and the economic context. Multipliers are higher when a country's current capital stock is low (Lama et al. (2019)) or when monetary policy is close to the zero lower bound. Absorptive capacity can also place a constraint on the efficiency or large investment scale-up (Adler (1965), Presbitero (2016)). In addition, recent research suggests that multipliers can be higher for green investments for renewable energy projects [1.1 – 1.7] compared to [0.4 – 0.7] for fossil fuel projects (Batini et al. (2021)), confirming the importance of adequate project screening and benefits from diversification away from the oil sector. By definition given the diversification aims, most NIS investments are not

Figure 7: Gross Fixed Capital Formation

(Percent of GDP)



directed towards fossil fuel projects.

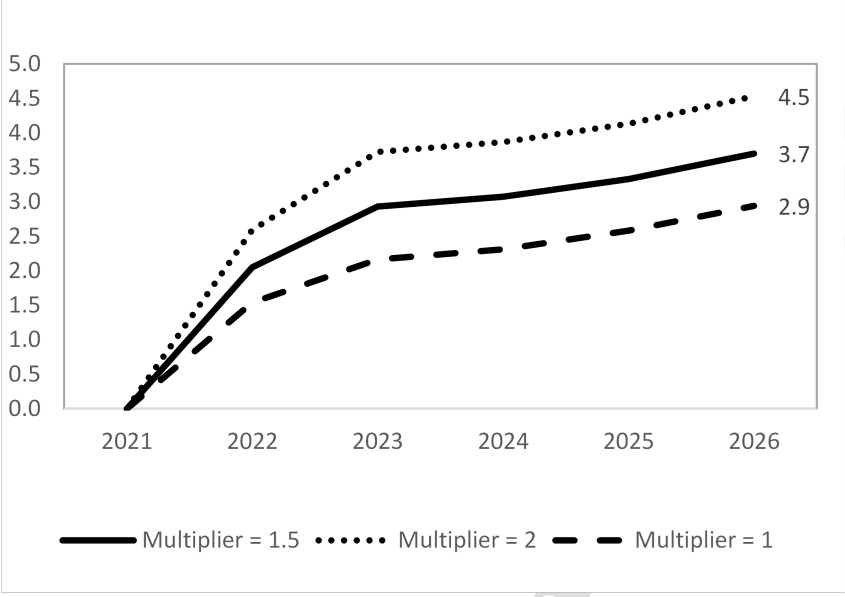
7 Robustness

To improve the robustness of the findings, we consider a range of multipliers. Investment multipliers are endogenous in the DIGNAR model, as they depend both on characteristics of the Saudi economy and estimated macroeconomic relationships. The main channels in the model include public investment efficiency, reallocation across sectors, adjustment costs, and the response of the private sector. In our preferred calibration, the model leads to a multiplier of 1.5, which sits at the middle of the range of public investment multipliers surveyed in Foster et al. (2023). Given the uncertainty, we also consider alternative specifications leading to multipliers of 1 and 2, which provide a range for the impact of the reform package and how large a catalytic role can be played by the PIF or through Shareek programs.

Alternative specifications continue to support a sizeable effect on non-oil potential growth. Setting the public investment multiplier at 1 – that is, one riyal in investment delivers one additional riyal in output in the medium-run – delivers an

Figure 8: Impact on Private investment

(Percentage points, relative to no-reform scenario)



additional boost of 2.9 percentage point in the reform scenario compared to the baseline (Figure 5). In contrast, if the multiplier is twice larger, the boost could reach 4.5 percentage points.

Another important source of uncertainty is the trajectory of oil prices, and their consequences on the Saudi economy, public finance, and the ability to stick to the investment plans. While no specific commitments have been made to direct additional fiscal resources from the central government to the PIF, authorities expect the PIF's assets under management to grow from around SAR 1.5 Tr currently, or 50 percent of GDP, to SAR 4.5 Tr in 2025, or about 100 percent of GDP. Such a growth of assets under management, if generated organically, would require PIF to consistently deliver returns twice the size of the economy's growth rate. Alternatively, returns on investment in the range commonly observed would fall short of the announced target for PIF's assets under management. As a result, in order to properly assess the future path of the PIF's assets under management and available investment capacity in both upside and downside scenarios, more details on the PIF financial relationship with the central government and debt issuance plans would

Journal Pre-proof

be required.[8]

Last but not least, geopolitical risks, including armed conflicts, are an important source of uncertainty in the region affecting medium-term growth. Compared to initial pre-conflict forecasts, actual real GDP per capita turned out nearly 5 percent lower (median) in the first year of an acute and prolonged conflict (International Monetary Fund, 2024). These risks can also play an important role in slowing down the diversification process in the Saudi Arabia, as suggested by a review of diversification efforts overt the past four decades (Sweidan and Elbargathi, 2023), and, more broadly, weigh down on financial returns among Gulf Cooperation Council (GCC) countries (Alqahtani et al., 2022).

8 Conclusion

We proposed to evaluate Saudi Arabia's growth strategy using an otherwise standard dynamic, general equilibrium framework. Through the lens of the model the strategy could deliver substantial growth, albeit slightly less that currently envisaged by authorities. In particular, our baseline scenario shows that by adhering to the NIS's investment plan, non-oil potential growth could increase by as much as 3.7 percentage points. Additionally, we assess alternative scenarios that vary in investment levels and fiscal multipliers. A key variable in our analysis remains the future direction of oil prices.

As projects are delivered, careful evaluation will allow to better assess the accuracy of the economic assumptions supporting the NIS. It will be important to assess the implicit multipliers associated with the various type of projects and account for uncertainty, both from external and internal factors. Furthermore, these average multipliers are likely to exhibit substantial heterogeneity across projects, with Giga-projects carrying a high risk/ high return profile. Careful monitoring of

[8]For details on the implications of oil prices on public finances, see IMF (2021) for the Selected Issue Paper on the fiscal anchor. PIF has received its debut A1 credit ratings in February 2022 by Moody's and green bonds were issued in October 2022 and February 2023 amounting to more than USD 8 billion.

projects through the PIF's Risk Committee and the NDF's new Impact Assessment Methodology will play an important role in maximizing medium-term risk-adjusted returns.

Policies to support innovation in key sectors could further lift potential output at longer horizons. While our simulations already find substantial gains, directing investments in R&D and technological upscaling in strategic sectors such as hydrogen or electric vehicles, may generate upside risks in the long run and complement improvements in the higher education system and human capital. Towards such a goal, a national plan for Research, Development, and Innovation (RDI) was unveiled in July 2022, with a target of annual spending of 2.5 percent of GDP on research and development by 2040, with priorities including health, energy, and environmental sustainability. Finally, an important avenue for research would be to how inclusive Saudia Arabia's economic transformation is of various gender, age, ethnic, or socio-economic groups.

References

Adler, J. H. (1965): *Absorptive capacity: The concept and its determinants*, Brookings Institution.

Al Naimi, S. M. (2022): "Economic Diversification Trends in the Gulf: The Case of Saudi Arabia," *Circular Economy and Sustainability*, 2, 221–230.

Al Rajhi, A., A. Al Salamah, M. Malik, and R. Wilson (2012): *Economic Development in Saudi Arabia*, Durham Modern Middle East and Islamic World Series, Routledge.

Albassam, B. A. (2015): "Economic diversification in Saudi Arabia: Myth or reality?" *Resources Policy*, 44, 112–117.

Aligishiev, Z., G. Melina, and L. F. Zanna (2021): "DIGNAR-19 Toolkit

Manual," IMF Technical Notes and Manuals 2021/007, International Monetary Fund.

Alqahtani, A., S. Hammoudeh, and R. Selmi (2022): "Relationship Between Different Sources of Geopolitical Risks and Stock Markets in the GCC Region: A Dynamic Correlation Analysis," *Review of Behavioral Finance*, 14, 296–3164.

Baqaee, D. R. and E. Farhi (2019): "The macroeconomic impact of microeconomic shocks: Beyond Hulten's theorem," *Econometrica*, 87, 1155–1203.

Batini, N., M. di Serio, M. Fragetta, G. Melina, and A. Waldron (2021): "Building Back Better: How Big are Green Spending Multipliers?" IMF Working Paper 2021/087, International Monetary Fund.

Baum, A., T. Mogues, and G. Verdier (2020): "Getting the Most from Public Investment," in *Infrastructure Governance: From Aspiration to Action*, ed. by G. Schwartz, M. Fouad, T. S. Hansen, and G. Verdier, Washington, DC: International Monetary Fund.

Berg, A., R. Portillo, S. C. S. Yang, and L. F. Zanna (2013): "Public investment in resource-abundant developing countries," *IMF Economic Review*, 61, 92–129.

Blazquez, J., M. Galeotti, B. Manzano, A. Pierru, and S. Pradhan (2021): "Effects of Saudi Arabia's economic reforms: Insights from a DSGE model," *Economic Modelling*, 95, 145–169.

Carvalho, V. M. and A. Tahbaz-Salehi (2019): "Production Networks: A Primer," *Annual Review of Economics*, 11, 635–663.

Cherif, R. and F. Hasanov (2016): "Soaring of the Gulf Falcons: Diversification in GCC Oil Exporters in Seven Propositions," in *Breaking the Oil Spell: The Gulf Falcons' Path to Diversification*, ed. by R. Cherif, F. Hasanov, and M. Zhu, Washington, DC: International Monetary Fund Press.

Journal Pre-proof

——— (2019): "The Leap of the Tiger: Escaping the Middle-income Trap to the Technological Frontier," *Global Policy*, 10, 497–511.

Cherif, R., F. Hasanov, and M. Sarsenbayev (2024): "Call of Duty: Industrial Policy for the Post-Oil Era," IMF Working Paper 24/74, International Monetary Fund.

Delechat, M. C., M. J. Clark JR, P. Gupta, M. M. Kabedi-Mbuyi, M. M. Koulet-Vickot, M. C. Macario, M. T. Orav, M. M. R. Torres, R. Tapsoba, D. Zhdankin, et al. (2015): *Harnessing resource wealth for inclusive growth in fragile states*, International Monetary Fund.

Foster, V., N. Gorgulu, S. Straub, and M. Vagliasindi (2023): "The Impact of Infrastructure on Development Outcomes," Policy Research Working Paper Series 10350, The World Bank.

Guendouz, A. A. and S. M. Ouassaf (2020): "The economic diversification in Saudi Arabia under the strategic vision 2030," *Acad. Account. Financ. Stud. J*, 24, 1–23.

Gurara, D., G. Melina, and L. F. Zanna (2019): "Some Policy Lessons from Country Applications of the Dig and DIGNAR Models," IMF Working Paper 19/62, International Monetary Fund.

Hausmann, R., L. Pritchett, and D. Rodrik (2005): "Growth Accelerations," *Journal of Economic Growth*, 10, 303–329.

Hvdit, M. (2013): "Economic Diversification in GCC Countries: Past Record and Future Trends," Research Paper Kuwait Programme on Development, Governance and Globalisation in the Gulf States, London School of Economics and Political Science, London, UK.

International Monetary Fund (2024): "Chapter 2: Fragile Foundations: The Lasting Economic Scars of Conflict," *Regional Economic Outlook, Middle East and Central Asia Department*.

Journal Pre-proof

Jawadi, F. and Z. Ftiti (2019): "Oil price collapse and challenges to economic transformation of Saudi Arabia: A time-series analysis," *Energy Economics*, 80, 12–19.

Lama, R., G. J. Vuletin, J. P. Medina, J. Puig, D. Riera-Crichton, C. A. Végh Gramont, and A. Izquierdo (2019): "Is the Public Investment Multiplier Higher in Developing Countries? An Empirical Exploration," IMF Working Papers 2019/289, International Monetary Fund.

Liu, E. (2019): "Industrial policies in production networks," *The Quarterly Journal of Economics*, 134, 1883–1948.

Manzano, O. and R. Rigobon (2007): "Resource curse or debt overhang," in *Natural Resources: Neither Curse nor Destiny*, ed. by D. Lederman and W. F. Maloney, Washington, D.C.: The World Bank.

Melina, G., S. C. S. Yang, and L. F. Zanna (2016): "Debt Sustainability, Public Investment and Natural Resources in Developing Countries: The DIGNAR Model," *Economic Modelling*, 52, 630–649.

Ministry of Investment (2022): "National Investment Strategy: Unleashing the Kingdom's Full Investment Potential," Accessed June 2022.

Pierru, A. and W. Matar (2014): "The Impact of Oil Price Volatility on Welfare in the Kingdom of Saudi Arabia: Implications for Public Investment Decision-making," *The Energy Journal*, 35, 97–116.

Prasad, A., K. Subramani, S. Refass, N. Saidi, F. Salem, and B. Shepherd (2024): "Global Economic Diversification Index 2024," Dubai: Mohammed bin Rashid School of Government, available at www.EconomicDiversification.com.

Presbitero, A. (2016): "Too much and too fast? Public investment scaling-up and absorptive capacity," *Journal of Development Economics*, 120, 17–31.

Journal Pre-proof

Sachs, J. D. and A. Warner (1995): "Natural resource abundance and economic growth," *National Bureau of Economic Research Working Paper 5398.*

Samargandi, N., J. Fidrmuc, and S. Ghosh (2014): "Financial development and economic growth in an oil-rich economy: The case of Saudi Arabia," *Economic modelling*, 43, 267–278.

Sweidan, O. D. (2021): "The Environmental and Energy Policies to Enable Sustainable Consumption and Production in the Gulf Cooperation Council Countries," *Clean Technologies and Environmental Policy*, 23, 2639–2654.

Sweidan, O. D. and K. Elbargathi (2023): "Economic diversification in Saudi Arabia: Comparing the impact of oil prices, geopolitical risk, and government expenditures," *International Economics*, 175, 13–24.

Vacher, J. (2022): "The need for a fiscal anchor and potential fiscal rules in Saudi Arabia," Article IV Staff Report on Saudi Arabia, Selected Issues, International Monetary Fund.

van der Ploeg, F. (2011): "Natural resources: curse or blessing?" *Journal of Economic Literature*, 49, 366–420.

——— (2012): "Bottlenecks in ramping up public investment," *International Tax and Public Finance*, 19, 509–538.

Journal Pre-proof

Journal Pre-proof

Declaration of competing interest: The authors do not have any conflict of interest.

Journal Pre-proof